

Analytical document - complete updated version

The DCEN Model v4

Democratic Cell Economic Network

- The 7 founding building blocks of the model
- Addition 1: consumer cooperatives that plan supply
- Addition 2: 0% interest investment for propagation
- References, precedents and honest caveats
- What is evidenced, what remains hypothetical

Analytical version: each statement distinguishes the documented fact from the hypothesis.
Inspired by Piketty, Saez, Zucman, EducationLibre, Mondragon, Migros/Coop, Raiffeisen.

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1. Overview - the model on one page

The DCEN model (Democratic Cell Economic Network) is an economic-transformation architecture that combines a diagnosis of inequality (Piketty, Saez, Zucman), a cooperative productive structure (SSE), decentralised taxation inspired by Swiss federalism, and peer-to-peer education (EducationLibre). This v4 version adds two elements that complete the architecture: consumer cooperatives that plan supply, and 0% interest investment that enables the model to propagate.

The logic in one sentence

We make wealth visible (blockchain), cantons tax it and pass it on in three shares, the sovereign fund invests in both production AND consumption cooperatives that coordinate democratically, the whole thing governed locally by citizens who train each other - and once the loop is established, the surplus finances, at 0% interest, the creation of identical cells elsewhere.

The 9 layers of the complete model

Layer	Element	Function
0 - Diagnosis	Piketty / Saez / Zucman	Document inequality and evasion
1 - Infrastructure	Global blockchain	Makes wealth visible and taxable
2 - Collection	Reversed cantonal taxation	Cantons collect, 40/35/25 key
3 - Capitalisation	Citizens sovereign fund	Invests in the real economy
4a - Production	Worker cooperatives	What we make and how
4b - Consumption	Consumer cooperatives	Plan the supply chain (NEW)
5 - Health / time	CCHI + 4-day week	Cooperative health, inclusion, freed time
6 - Coordination	Democratic coordination	Federated cells, market as signal
7 - Governance	Swiss model	Subsidiarity, municipalities to confederation
8 - Knowledge	P2P education (EducationLibre)	Autonomous citizens, reciprocity credits
9 - Propagation	0% interest investment	Creates cells elsewhere (NEW)

2. The 7 founding building blocks (synthetic recap)

These blocks were documented in detail in previous reports. Here is a synthetic recap to situate the two additions in their context.

Global wealth blockchain

A universal register making wealth observable, therefore taxable. A logical extension of Zucman's work on traceability. To be distinguished from a cryptocurrency: it is a fiscal transparency infrastructure, not a speculative asset.

Reversed cantonal taxation (40/35/25 key)

Cantons collect the progressive tax and pass it on: 40% stays with the canton, 35% to the sovereign fund, 25% to equalisation. An inversion of the usual fiscal chain, consistent with existing Swiss federalism.

Citizens sovereign fund

Public capitalisation with democratic governance, investing in the real economy rather than speculating. Reference models: Norway, Alaska, Singapore.

SSE - production cooperatives

Worker cooperatives, mutuals, associations with shared governance. This is the production side of the cooperative economy.

CCHI - cooperative health

Cooperative health-insurance funds replacing the private part of Swiss KVG, with income-proportional contributions. Connects directly to consumer pharmacy cooperatives (addition 1).

Reconfigured work

A 4-day week and active inclusion of the unemployed and sick into cooperative structures. The freed time makes citizen governance and P2P learning possible.

P2P education (EducationLibre)

A peer-to-peer learning system inspired by BitTorrent, with reciprocity credits NOT convertible into money. Makes citizens able to understand and govern the system.

3. Addition 1 - Consumer cooperatives

Until now, the DCEN model dealt mainly with production (who makes, how). But one question remained without a clear answer: what signal tells the production cooperatives what to produce? The answer is consumer cooperatives.

3.1 The principle: planning by demand, not by supply

Rather than a central state planning production (the historical error of the Soviet Gosplan), it is the consumer cooperatives - supermarkets, pharmacies - that plan their supply. They aggregate people's real demand, as close to the ground as possible. They do not have to guess needs: they observe them directly in what their members buy.

NEW

The economic calculation problem is solved by the right actor

Hayek and von Mises were right: a centre cannot know the dispersed needs of each locality. But consumer cooperatives are in direct contact with the need. They hold the local information that the centre could not aggregate.

NEW

The pharmacy case is particularly relevant

Medicine is a good where the pure market often fails: recurrent shortages, opaque prices, dependence on a few suppliers. A pharmacy cooperative planning its supply at canton scale could pool stocks, smooth shortages, negotiate prices collectively, and prioritise according to real health needs. It connects directly to the CCHI (cooperative health).

NEW

Aggregated demand orients production

This is the missing link: the signal that tells production cooperatives what to make comes neither from a state plan nor from the market alone - it comes from consumer cooperatives that democratically relay real needs from the ground up.

3.2 Why Switzerland is the ideal ground

This is not a utopia: Switzerland has already done this at national scale, and these structures dominate its economy. Migros and Coop are consumer cooperatives that effectively plan their supply chains, including by integrating their own production. The model therefore proposes to generalise and democratise an already-proven model, extending it to other sectors such as pharmacy.

4. Addition 2 - 0% interest investment

Once the core of the DCEN model is operational and generating a surplus, how does the model propagate? Not through imposition, not through predatory debt - through 0% interest investment.

4.1 The principle: propagate, not extract

When the IMF or the World Bank lend to a country, they extract interest and impose conditions (austerity, privatisations). The DCEN sovereign fund does the opposite: it lends without interest, for the sole purpose of having the model reproduce elsewhere. This is not naive philanthropy - it is a calculation: each DCEN cell that succeeds elsewhere strengthens the legitimacy and resilience of the whole network.

NEW

Three concentric circles of propagation

First neighbouring cantons (cultural proximity, the easiest: Jura, Vaud, Bern). Then border regions (France, Italy, Germany). Then the Global South, where the absence of predatory debt is a decisive argument against Chinese or Western competition.

NEW

Each cell becomes an investor in turn

This is the viral propagation mechanism. A cell that succeeds does not stay dependent: it generates its own surplus and finances the next cell in turn. This is exactly how the Raiffeisen banks multiplied in the 19th century, or how Mondragon spread.

NEW

Propagation by example, not by imposition

Since the model is open source (like EducationLibre, under a free licence), any region can adopt and adapt it to its context. Diffusion is organic: no one is forced, we show that it works and finance those who want to try.

4.2 The critical condition: a real and durable surplus

WARNING - essential caveat: 0% investment assumes that the sovereign fund generates a real and durable surplus. As long as the core of the model has not demonstrated that it produces this surplus, this propagation layer remains purely conditional. We cannot start by exporting: first a first cell must prove it works and generates a surplus. The sequence is non-negotiable.

5. How it all fits together - the double loop

The v4 model works through two nested loops: an internal loop (production-consumption) and an external loop (propagation).

5.1 The internal loop: production - consumption

Consumer cooperatives observe real demand and plan their supply. This aggregated demand flows up to production cooperatives, which adjust what they make. The market remains an information signal (prices indicate scarcities), but it is no longer the master: the final decision belongs to the cells, democratically. This can be called democratic productive coordination, rather than central planning.

5.2 The external loop: propagation through surplus

Once the internal loop is stabilised, the sovereign fund generates a surplus. This surplus finances, at 0% interest, the creation of DCEN cells elsewhere. Each new cell, once mature, generates its own surplus and finances the next. The growing network of cells reinforces, in return, the original core (more partners, more legitimacy, more resilience). It is mutual reinforcement, not one-way extraction.

5.3 Why this articulation is coherent

The same principle at two scales

At the local scale, consumer cooperatives orient production through demand. At the continental scale, mature cells orient the creation of new cells through 0% investment. In both cases: information and decision stay as close to the ground as possible, and solidarity circulates without extraction.

6. Real precedents and references

Each of the two additions rests on documented precedents. Here are the facts supported by verifiable sources.

6.1 Consumer cooperatives: Migros, Coop

EVIDENCED

Consumer cooperatives already dominate Swiss retail

Coop was born in 1864 as a consumers' cooperative. Migros was transformed into a cooperative by Gottlieb Duttweiler in 1941, who donated it to its customers. Together they represent a major share of Swiss retail and effectively plan their supply chains, even integrating their own production. The DCEN model proposes to democratise and extend what already exists.

Sources: Coop Switzerland (history, founded 1864); Migros / Federation of Migros Cooperatives (cooperative transformation 1941, Gottlieb Duttweiler). Public data on Swiss retail market shares.

6.2 Cooperative propagation: Raiffeisen, Mondragon

EVIDENCED

Raiffeisen: propagation by spreading in the 19th century

The Raiffeisen banking model, founded on cooperation and mutual aid, spread from village to village from the mid-19th century. In Switzerland, Raiffeisen is today one of the main banking groups, run by its members - proof that the cooperative model can spread and endure.

EVIDENCED

Mondragon: 70,000 jobs from 5 workers

Founded in 1956 by the priest Arizmendiarieta with his students, the Mondragon Corporation employed 70,085 people in 2024 (revenue of 11.2 billion euros). Of the 103 cooperatives created between 1956 and 1986, only three closed (a survival rate of about 97% over three decades). Mondragon spread by reinvesting its surpluses - the exact mechanism that 0% investment proposes.

Sources: Wikipedia / Mondragon Corporation Annual Report 2024; Participedia; Corporate Rebels (2024); history of the Raiffeisen movement (Friedrich Wilhelm Raiffeisen, from the 1860s).

7. Honest caveats and limits

Intellectual integrity requires naming the difficulties of the two additions, not just their promises.

7.1 Limits of consumer cooperatives

CAVEAT

Upstream dependence does not disappear

A consumer cooperative plans its demand, but depends on what producers supply. For medicines, a Swiss pharmacy cooperative remains dependent on the global pharmaceutical industry. Demand planning cushions this dependence, it does not eliminate it - unless one moves upstream in the chain (cooperative production of generics).

CAVEAT

The critical-mass risk

Migros and Coop are powerful because they are enormous - their negotiating power comes from volume. A small local cooperative has little weight against suppliers. Hence the importance of the network logic: autonomous but federated local cooperatives that carry weight together (cooperative purchasing centrals).

CAVEAT

Real democratic governance, not formal

Migros and Coop are today criticised because, having become giants, their cooperative dimension is more formal than lived - few members truly participate. The model must avoid this pitfall through P2P education and proximity governance.

7.2 Limits of 0% investment

CAVEAT

Everything depends on a surplus that does not yet exist

This is the most important limit: without a real and durable surplus from the sovereign fund, propagation is impossible. And this surplus is today only a hypothesis - the core of the model has never been tested at real scale. Propagation is therefore a conditional promise, not an established fact.

CAVEAT

The risk of paternalism

Even at 0% interest, exporting a model to other regions can be perceived as interference or paternalism. The open-source character and free (non-imposed) adoption are safeguards, but vigilance remains necessary.

CAVEAT

Long-term sustainability

Lending without interest ties up capital. If too many cells fail or take too long to become autonomous, the source fund can be depleted. The pace of propagation must be cautious and indexed to the real maturation of existing cells.

8. Implementation sequence

Order matters. We cannot do everything at once. Here is a coherent sequence, from the most immediate to the most distant.

Step	What we build	Condition to proceed	Timeline
1	Pilot cooperative (production AND consumption) in a municipality	Results measured over 18-36 months	Immediate
2	Pharmacy cooperative linked to an embryonic CCHI	Aggregated local demand functional	Year 2-4
3	Democratic coordination between cells in the same canton	Several federated cooperatives	Year 3-6
4	Cantonal sovereign fund generating a first surplus	Internal loop stabilised and in surplus	Year 5-10
5	First 0% investment to a neighbouring canton	Real and durable surplus confirmed	Year 8-12
6	Network of cells - cross-border propagation	Several autonomous, surplus-generating cells	Year 12-20

The guiding principle of the sequence

Each step must prove its value before the next. We do not propagate a model that has not proven itself, we do not claim at 0% a surplus that does not exist, we do not federate cooperatives that do not yet exist. This is exactly the small-steps method that Jean Monnet applied to European construction: concrete institutions that prove their value before extending. The difference: here, the method is democratic from end to end.

SOURCES - Evidenced facts: Coop Switzerland (founded 1864); Federation of Migros Cooperatives (transformation 1941, G. Duttweiler); Mondragon Corporation Annual Report 2024; Raiffeisen movement (F. W. Raiffeisen, 1860s); G. Zucman, The Hidden Wealth of Nations (2015) and G20 report (2024); F. Hayek, The Use of Knowledge in Society (1945) on the economic calculation problem. Modelling assumptions: the production-consumption articulation and the 0% propagation mechanism are theoretical constructions of this work, never tested in this combined form. EducationLibre: github.com/nadnone/EducationLibre (CC BY 4.0).